International Emergency Nursing

# Exploring the impact of experiencing a long lie fall on physical and clinical outcomes in older people requiring an ambulance: a systematic review.

--Manuscript Draft--

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| **Manuscript Number:** | YAAEN-D-21-00221R1 |
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| **Abstract:** | Abstract<br>Background: The long term impacts of experiencing a 'long lie' following a fall in the older person are poorly understood. This systematic review explored the impact of a long lie fall on physical and clinical outcomes in older people requiring an ambulance.<br>Methods: PRISMA guidelines were followed.<br>Results: 70 studies were identified. Nine studies were suitable for full review. Four studies meeting the inclusion criteria were included. The Critical Appraisal Skill Programme (CASP) assessed the quality of all included studies. Three studies reported on people aged 65 years and older. One study reported on people aged over 90 years. Personal alarm use was examined in two studies. One study explored patient characteristics of people confirmed to have fallen by paramedics at the scene. One study examined re-contact and characteristics of fallers referred to a falls prevention service.<br>Discussion: Cognitive impairment and long lie were a caveat for falls and repeated falls. Personal alarm use was infrequent suggesting a need for supporting the older patient in appropriate alarm use and exploration of newer technologies to alleviate their need. Future research should focus on interventions for wearable, smart and e-technology for automatic fall detection and qualitative exploration of the lived experience. |
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Title page with Author details

Title: Exploring the impact of experiencing a long lie fall on physical and clinical outcomes in older people requiring an ambulance: a systematic review.

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Keywords

Accidental falls; frailty; basic ambulance care; admission avoidance; first responders

Abstract

**Background:** The long term impacts of experiencing a ‘long lie’ following a fall in the older person are poorly understood. This systematic review explored the impact of a long lie fall on physical and clinical outcomes in older people requiring an ambulance.

**Methods:** PRISMA guidelines were followed.

**Results:** 70 studies were identified. Nine studies were suitable for full review. Four studies meeting the inclusion criteria were included. The Critical Appraisal Skill Programme (CASP) assessed the quality of all included studies. Three studies reported on people aged 65 years and older. One study reported on people aged over 90 years. Personal alarm use was examined in two studies. One study explored patient characteristics of people confirmed to have fallen by paramedics at the scene. One study examined re-contact and characteristics of fallers referred to a falls prevention service.

**Discussion:** Cognitive impairment and long lie were a caveat for falls and repeated falls. Personal alarm use was infrequent suggesting a need for supporting the older patient in appropriate alarm use and exploration of newer technologies to alleviate their need. Future research should focus on interventions for wearable, smart and e-technology for automatic fall detection and qualitative exploration of the lived experience.

Key Words

Long lie; Falls; Elderly; Ambulance response; Fall detection technology; Lived experience; Patient experience



# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | Title |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | Abstract |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | Page 3 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | Page 3 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | Page 4 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | Page 4 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | Page 4 |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | Page 4 |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | Page 4 |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | Risk of bias table |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | Risk of bias table |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | Risk of bias table |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | Risk of bias table |
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | |
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | |
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | Risk of bias table |
| Certainty | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | Risk of bias |



# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| assessment | | | table |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | Page 13 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | Page 13 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | Page 13 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | Risk of bias table |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Risk of bias table |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | |
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | Risk of bias table |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | Page 15 |
| | 23b | Discuss any limitations of the evidence included in the review. | Page 15 |
| | 23c | Discuss any limitations of the review processes used. | Page 15 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | Page 15 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | Page 26 |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | Page 26 |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | Page 26 |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | Page 26 |
| Competing interests | 26 | Declare any competing interests of review authors. | Page 26 |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | Page 26 |

*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

# PRISMA 2020 Checklist

For more information, visit: http://www.prisma-statement.org/

# Highlights

What is already known on this subject?

- The long term physical and clinical outcomes for older people requiring an ambulance response following a long lie are poorly understood.
- There is limited evidence on the association between long lie and physical and clinical outcomes in older people.
- A reluctance for personal alarm use demonstrates a need for supporting patients on appropriate alarm activation.
- Research into evolving interventions for wearable, smart and e-technology for automatic fall detection is needed.

Click here to access/download;Table;Figure 1_PRISMA_revision_final_310321.pdf

**Figure 1: PRISMA Flow Diagram**

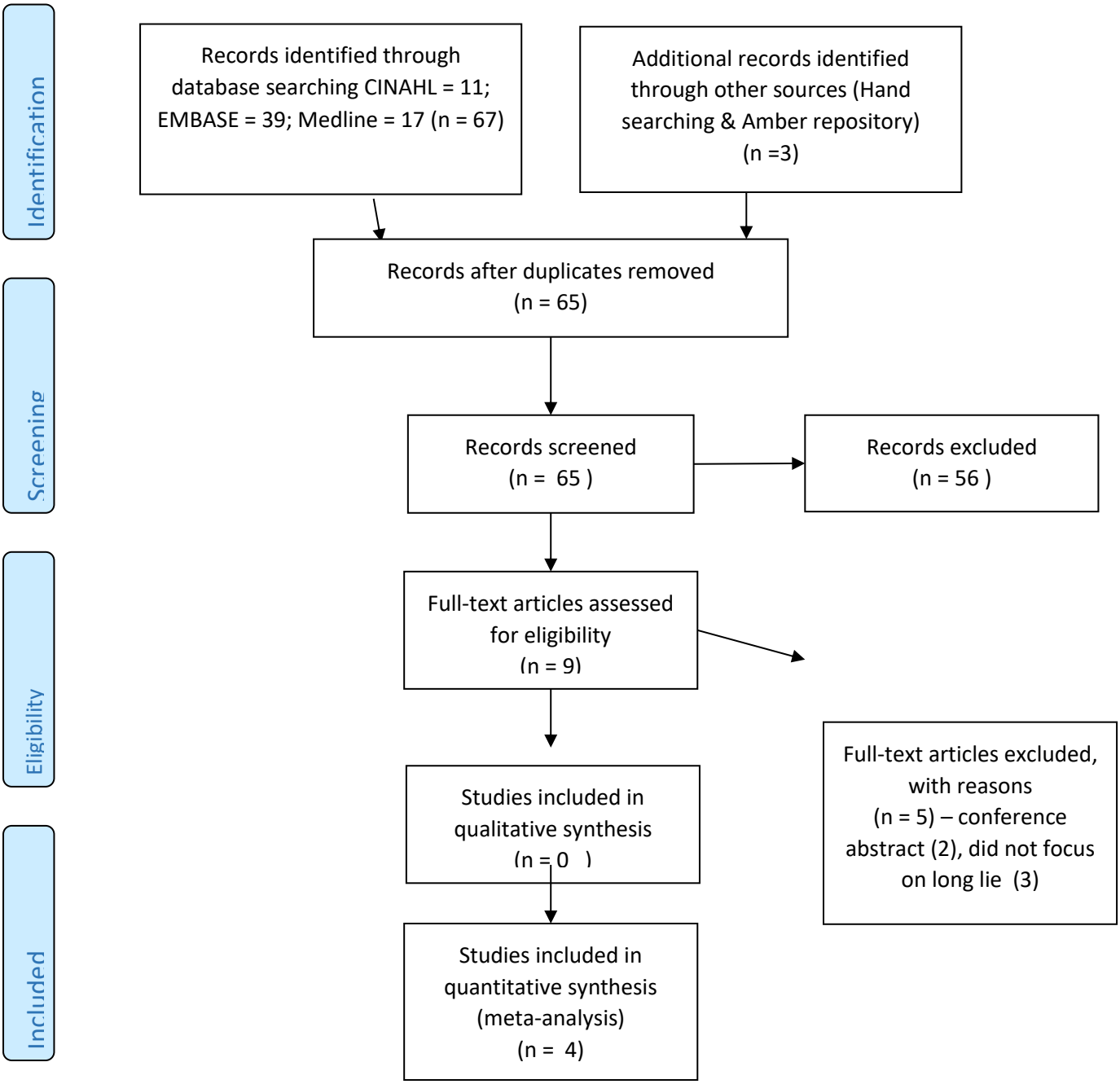


Table 1: Risk of bias

Click here to access/download;Table;Table 1_Risk of bias of studies_270921.docx

| Study | Potential source of bias | Study parameters | Assessed risk of bias |
|---|---|---|---|
| Scott (2020) | Was selection of exposed cohorts drawn from the same population? | All individuals aged ≥ 65 years in the geographical area covered by the South Eastern division of the Northern Ireland Ambulance Service (NIAS) who were attended by an NIAS paramedic for a fall and referred to the falls prevention service during the 3-month period from 1 July to 30 September 2017 were included. | None |
| | Can we be confident in the assessment of exposure? | Participants were identified using NIAS Emergency Ambulance Control (EAC) call data for incidents coded by the Advanced Medical Priority Dispatch System (AMPDS) as 'falls' and/or those whose main complaint recorded on Patient Report Forms (PRFs) by the ambulance crew was a fall. Re-contacts were identified by searching the EAC database for subsequent contacts with the NIAS within 1-month and 6-month follow-up. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – the cohort looked at re-contacts with the ambulance service so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Outcomes were measured using PRF data and the EAC control database. Clinical data, including medical history, drug history, clinical observations, treatments and onset time of the fall, were collected from PRF data. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | Yes - outcomes were measured using PRF data and the EAC control database. Clinical data, including medical history, drug history, clinical observations, treatments and onset time of the fall, were collected from PRF data. | Very low |
| | Can we be confident in the assessment of outcome? | Yes - the primary outcome was the rate of subsequent contacts with the ambulance service following referral. There were 1079 incidents of falls in service users aged ≥ 65 years. A referral rate of 7% (n = 75) was reported, | Very low |

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|---|---|---|---|
| | | constituting the study cohort. Re-contact rates were 37.3% (n = 28) within 1 month and 70.7% (n = 53) within 6 months. | |
| | Was the follow-up of cohorts adequate? | Participants were followed up for 6 months. | Low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Fleming & Brayne (2008) | Was selection of exposed cohorts drawn from the same population? | Both groups comprised surviving participants drawn from the same cohort; the Cambridge City over-75s Cohort, a population based sample. | None |
| | Can we be confident in the assessment of exposure? | Falls data was obtained when reported by telephone, via a weekly falls report or follow-up visit or phone call by study nurse. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Yes - odds ratios and hazard ratios were adjusted for age, sex, and mobility. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | The longitudinal cohort study of older old people initially recruited participants through general practices in the 1980s, when they were all aged 75 or over. Repeated surveys since baseline have gathered data on a range of variables including socio-demographics, physical and mental health, function, and detailed cognitive assessment that included the mini-mental state examination. | Very low |
| | Can we be confident in the assessment of outcome? | Data recorded after each fall included whether the individual who fell had been able to get up without help, how long they were on the floor, any injuries, and whether they called for assistance. | Very low |

| | | | |
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| | Was the follow-up of cohorts adequate? | Participants were followed-up intensively for 1 year. | Very low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Johnston et al. (2010) | Was selection of exposed cohorts drawn from the same population? | All data was obtained from database files provided by SAAS (South Australian Ambulance Service). | None |
| | Can we be confident in the assessment of exposure? | Excluding alarm activation error, all patients ≥ 65 years of age who used a personal alarm to call the ambulance were identified. | Low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – the primary outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Cases were matched by date of service, ambulance action (i.e. transport to hospital or lift only) and age with patients who did not use an alarm. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | A purpose-built data extraction file was used to record de-identified information extracted from each falls case including demographics: age, gender service details: date, type of service, response time fall details: mechanism, location, other persons present, location of person when ambulance arrived, reported injury from fall ambulance contact details: alarm use, who called ambulance fall risk factors: medical conditions, medications, cognitive status, balance/mobility issues, continence, falls history, visual/sensory problems, environmental factors. | Very low |
| | Can we be confident in the assessment of outcome? | The study outcomes were falls, or alert button activation error. These were identified by codes allocated when individuals activated a personal alarm, an ambulance was dispatched, but at the time of ambulance attendance the individual did not require either a lift, or transport to hospital. | Low |
| | Was the follow-up of cohorts adequate? | Participants were followed up for 1 month. | Low |

| | | | |
|---|---|---|---|
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Simpson et al. (2014) | Was selection of exposed cohorts drawn from the same population? | This study was a retrospective descriptive analysis of routinely collected data from the Ambulance Service of New South Wales (ASNSW), Australia.<br>Both cohorts were drawn from databases of the Ambulance Service of New South Wales (ASNSW). | None |
| | Can we be confident in the assessment of exposure? | Exposure is whether or not ambulance transport was required. Data contained dispatch and 'triple zero' call information from the Medical Priority Dispatch System (MPDS) software. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | A series of unadjusted analyses were conducted considering prognostic variables on a one-at-a-time basis. | Possible |
| | Can we be confident in the assessment of the presence of prognostic factors? | Data contained demographic and clinical information about the patient, including basic physiological measurements, identified injuries, and paramedic treatment. | Very low |
| | Can we be confident in the assessment of outcome? | MPDS includes a falls category. | Low |
| | Was the follow-up of cohorts adequate? | Participants were followed-up for 1 year. | Very low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |

*Table 1: Risk of bias of studies*

Click here to access/download;Table;Table 2_includesstudies_270921.docx

*Table 1: Table of included studies*

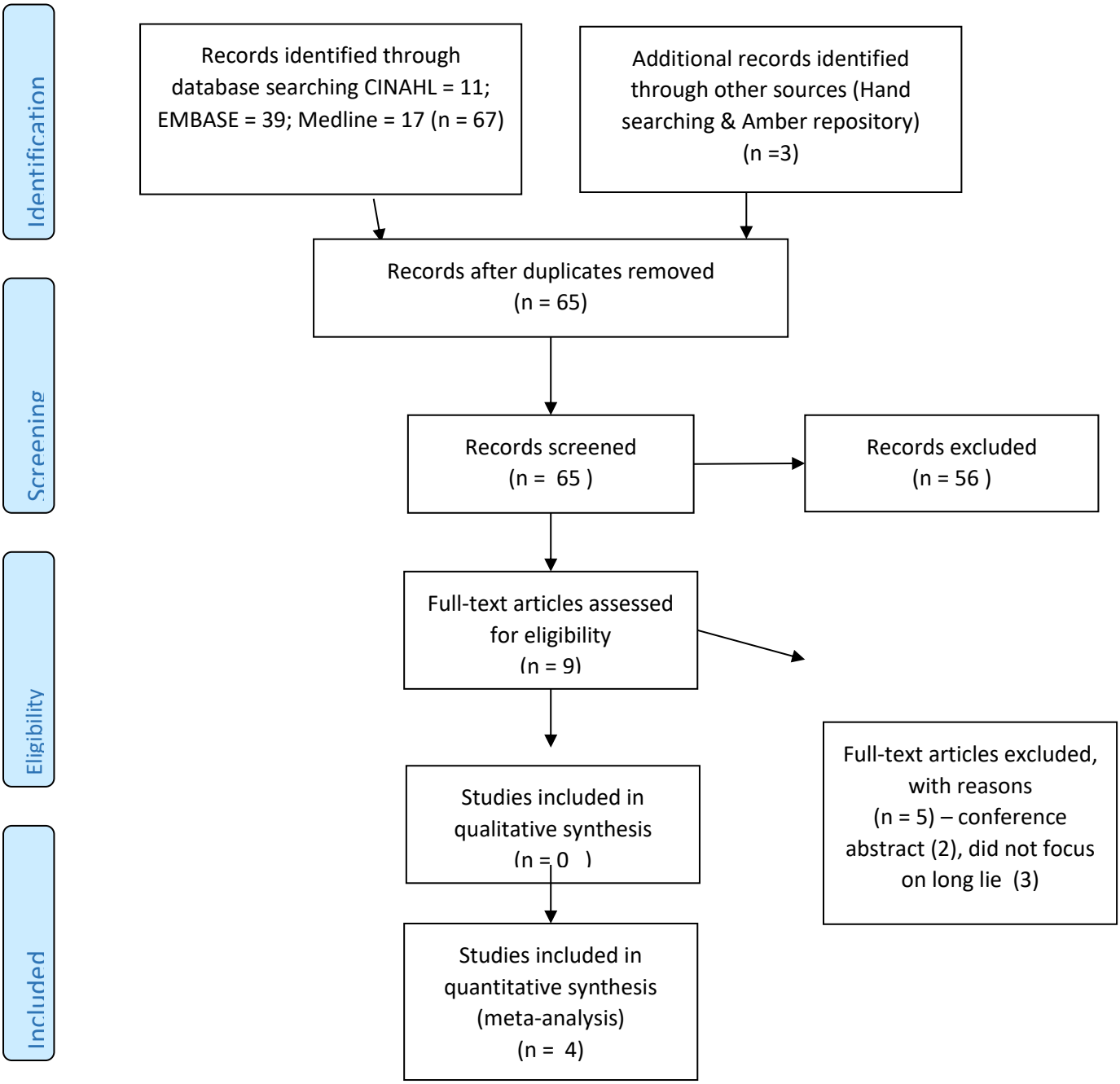
| **Author/year/ Country** | **Design/setting/ population details** | **Outcomes being assessed** | **Main findings** | **Conclusion** |
|---|---|---|---|---|
| Johnston, K., Worley, A., Grimmer-Somers, K., Sutherland, M., & Amos, L. (2010).<br><br>Australia | A retrospective one month audit of South Australian Ambulance Service records (n=1700) | Characteristics of ambulance call outs for falls<br><br>Use of personal alarms by the fallers | Ambulance response time did not differ between alarm and non-alarm users.<br><br>82% of alarm users and 75% of non-alarm users were on the ground when the ambulance arrived.<br><br>11 of non-alarm users self-reported 'long lies' (>one hour) before the ambulance was called<br><br>There were 13 other cases with unknown time on the floor. | Older women living alone were the major users of personal alarms for assistance after falling.<br><br>If activated quickly, alarms enabled most fallers to gain ambulance attention within 15 minutes.<br><br>Personal alarm use was associated with a high incidence of false alarms. |
| Fleming, J., & Brayne, C. (2008).<br><br>England | 1 year follow-up of participants in a prospective cohort study of ageing, using fall calendars, phone calls, and visits (n=110) | Inability to get up without help, lying on floor for a long time after falling, associated factors; availability and use of call alarm systems.<br><br>Participants' views on using call alarms | 54% of fall reports described the participant as being found on the floor and 82% of falls occurred when the person was alone.<br><br>Of the 60% who fell, 30% had lain on the floor for an hour or more.<br><br>Difficulty in getting up was consistently associated with age, reported mobility, and severe cognitive impairment. | This study identifies the oldest people are more at risk of falls and that the design of personal alarms requires development. Call alarms were widely available but were not used in most cases of falls. |

| | | | | |
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| | | to summon help if needed after falling. | Cognition was the only characteristic that predicted lying on the floor for a long time.<br><br>Lying on the floor for a long time was strongly associated with serious injuries, admission to hospital, and subsequent moves into long term care. | |
| Simpson, P. M., Bendall, J. C., Tiedemann, A., Lord, S. R., & Close, J. C. T. (2014). Australia | Prospective cohort study of people aged ≥65 who had fallen and called for an ambulance (n=1610) | Collection of fall-related data allowing for a sequential description of fall-related cases from time of ambulance dispatch through to the end of the prehospital episode of care. | From 1,610 eligible cases and "long-lies" (>60 minutes on the ground) occurred in 13% of cases.<br><br>Patients were predominantly female and community dwelling.<br><br>The most common patient-reported reasons for falling were:<br><br>Loss of balance<br><br>Simple trips<br><br>New injury and/or pain was documented for 1,172 of patients<br><br>656 presented with abnormal" physiology | The paper concludes ambulance services appear to provide timely responses to older people who have fallen, and "long-lies" are relatively uncommon. |

| | | | | |
|---|---|---|---|---|
| | | | Only 238 presented with no new injury/pain and normal physiology. | |
| Scott, J (2020) | Retrospective cross-sectional cohort study in South Eastern division of the Northern Ireland Ambulance Service (NIAS). | Subsequent contacts with the ambulance service following referral. | There were 1079 incidents of falls in service users aged ≥ 65 years with a referral rate of 7% (n = 75) was reported. Re-contact rates were 37.3% (n = 28) within 1 month and 70.7% (n = 53) within 6 months.<br><br>Women and those experiencing a 'long lie' were more likely to re-contact. Those with cognitive impairment were vulnerable to falls and repeat falls. Documentation for recording patient-related information was poor. | Interventions targeted at reducing long lies and supporting patients with cognitive impairments should be explored. Clinical documentation needs to be improved. |

Click here to access/download;Figure;Figure 1_PRISMA_270921.pdf

**Figure 1: PRISMA Flow Diagram**



A table of response to reviewers comments

| Reviewer comments | Response |
|---|---|
| Comments relating to the need to specify a country when referencing. | Thank you for this comment. Countries have been added to the relevant sections of the manuscript. |
| Adapted critical appraisal tools 18 were used to assess the papers – how and why were these adapted? | Thank you for this comment. The CASP tool is a generic appraisal tool. In order to answer the review, aim and objectives, additional questions were used to appraise and extract the information in order to systematically review all the included and excluded papers. |
| The quality of all included studies was assessed using appropriate appraisal checklist developed by the Critical Appraisal Skill Programme (CASP). 18 – This has already been stated | This has been removed from this section of the paper. |
| Understand - grammar | This has been changed to 'understood' |

Dear Editor.

Thank you for reviewing our manuscript and for giving us the opportunity to provide minor amendments.

Please find attached a revision to our manuscript entitled ‘Exploring the impact of experiencing a long lie fall on physical and clinical outcomes in older people requiring an ambulance: a systematic review.’

We hope our revision will be favourably considered and look forward to hearing from you in due course.

Many thanks

Joanna Blackburn

Clean copy of the manuscript (without Author details)

Click here to view linked References


## Abstract

**Background:** The long term impacts of experiencing a 'long lie' following a fall in the older person are poorly understood. This systematic review explored the impact of a long lie fall on physical and clinical outcomes in older people requiring an ambulance.

**Methods:** PRISMA guidelines were followed.

**Results:** 70 studies were identified. Nine studies were suitable for full review. Four studies meeting the inclusion criteria were included. The Critical Appraisal Skill Programme (CASP) assessed the quality of all included studies. Three studies reported on people aged 65 years and older. One study reported on people aged over 90 years. Personal alarm use was examined in two studies. One study explored patient characteristics of people confirmed to have fallen by paramedics at the scene. One study examined re-contact and characteristics of fallers referred to a falls prevention service.

**Discussion:** Cognitive impairment and long lie were a caveat for falls and repeated falls. Personal alarm use was infrequent suggesting a need for supporting the older patient in appropriate alarm use and exploration of newer technologies to alleviate their need. Future research should focus on interventions for wearable, smart and e-technology for automatic fall detection and qualitative exploration of the lived experience.



## Key Words

Long lie; Falls; Elderly; Ambulance response; Fall detection technology; Lived experience; Patient experience

# Introduction

Falls are a common occurrence and are the largest cause of hospital admission in the older person; between 2017-2018 approximately 220,160 emergency hospital admissions in the United Kingdom (UK) were attributable to falls in this patient group (aged 65 years and over) with just over 66% of these patients aged over 80 years. [1] Hospital-related falls account for approximately 240,000 reported in-patient safety incidents in UK hospitals [2], with an estimated cost of 2.3 billion. [3] A fall is defined as an event which causes a person to, unintentionally, rest on the ground or other lower level. For the purposes of this paper, a non-traumatic fall is defined as a fall not causing significant trauma or injury to the patient. [4]

The prevention and management of falls in older people have been the focus of UK national government initiatives which have endeavoured to focus on the reduction of morbidity and mortality in this patient group. [5, 6] The ambulance service is under growing pressure to provide care provision to an increasing number of older people who fall in their own home [7]; a frequent reason for an emergency ambulance call. [8] Even in those people who do not necessitate hospital admission, there is an increased likelihood of poorer health outcomes, an increased risk of future admission to hospital and increased morbidity than the general older person population. [9] Community related falls are largely preventable, however, data suggests that approximately 1 in 3 adults over 65 will experience one fall annually [3] and an estimated 5% of falls in the community result in hospitalisation for injuries such as fractures. [10] Snooks et al [11] found that 8% of all calls made to the London Ambulance Service were due to older people who had fallen at home, with 36% of these people not requiring an admission to hospital, despite 49% of fallers requiring healthcare assistance in the two-weeks post-fall, with an increased risk of death and hospital admission than the general population.

Clinical and physical age-related complications such as muscle weakness, visual impairment or loss of balance, as well as environmental factors such as a hazardous environment increase fall-related risk. Hazards in the home resulting in a fall cost the NHS in the UK approximately £435 million [12] and adapting the home environment to improve safety and limit fall-risk has been a focus of interventions in falls prevention. [13] Suffering a fall at home can be a precursor to further future falls and the rehabilitative resources required to manage an older person post-fall can place significant demands on healthcare services. [3]

**Experiencing a long lie fall**

A potentially serious complication of community-related falls in older people is the inability to get up following a fall, which is associated with severe psychological, emotional and physical distress connected to a fear of future falls, pressure-related injuries and loss of mobility. [3] Providing information on how to avoid a 'long lie', generally defined as being unable to get up from the floor for over one hour following a fall, is a priority in the NICE recommendations for preventing falls in older people. [3] However, there is scant evidence surrounding the long-term outcomes of people who experience a long lie fall in the community setting. Fleming et al [14] undertook a prospective cohort study in the UK to examine the use of call alarm systems and their association with experiencing a long lie in people over 90 years of age who had fallen at home. In a sample of 110 participants, at one-year follow-up 60% were reported to have experienced a fall on at least one occasion and falls in the community setting accounted for 45% of all reported falls. Interestingly, just over half of all reported falls resulted in the participant being found on the floor, with the majority of these participants requiring assistance to get up from the floor. A total of 15% of all reported falls resulted in a long lie, however, accounting for the individual people who fell rather than the reported number of falls, long lies represented 30% of falls. Those who did experience a long lie were more likely to suffer subsequent serious injuries, admissions to hospital and future long term care. Conversely, Simpson et al [15] explored the characteristics of older people who had fallen in Australia and called an ambulance through a prospective cohort study from a population of 1,610 cases and found that just over 53% of patients were still on the floor when the ambulance arrived. Due to the quick ambulance response time, only 13% of these patients experienced a "long-lie". Cannon et al [16] found no relationship between ambulance response time and long lie in a sample of 503 patients.

The physical and clinical impacts on the older person experiencing a long lie remain poorly understood with variations in the patient population investigated, study methodologies and a lack of robust clinical evidence contributing to a lack of understanding around the long term impacts of a long lie fall. As the ambulance service are required to respond to an increasing number of older people who fall, understanding the physical and clinical impacts of a fall is fundamental to limiting the number of people suffering a long lie.

## Aims and objectives

This systematic review aimed to explore the relationship between a non-traumatic fall in people aged 65 years or over resulting in a long lie (defined as remaining on the floor for an hour or longer) and subsequently cared for by the ambulance service, and the associated physical and clinical health related outcomes. The following objectives were defined to achieve this aim:

1. To determine if there is an association between length of time spent on the floor after falling and physical and clinical outcomes for older patients.

2. To understand whether the type and timeliness of response provided by the ambulance service has any physical or clinical consequences for older patients who have a long-lie after falling.

# Methods

The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) guidelines were followed (Figure 1). [17] Adapted critical appraisal tools [18] were used to assess the papers.

## Inclusion and exclusion criteria

The population of interest were patients over 65 years of age who suffered a non-traumatic fall and subsequent long lie (defined as remaining on the floor for >60 minutes), and who were cared for by the ambulance service. Randomised controlled trials (RCTs), cohort, observational studies whether prospective or retrospective in design, and audits were suitable for inclusion. No constraints were placed on language of publication or date of publication. Conference papers, poster abstracts, opinion papers and case studies were excluded.

## Search strategy.

PubMed, Embase, Medline, CINAHL, and the Amber repository were searched from 1980 to March 2021 using keywords and medical subject headings (mesh) determined by the authors: ("EMERGENCY MEDICAL SERVICES"/ OR AMBULANCES/ OR (paramedics).ti,ab) AND ((long).ti,ab AND (lie OR lay). The study team also reviewed the reference lists of retrieved studies to identify studies, which had not been identified by the search strategy. Duplicate studies were excluded.

## Quality assessment and data extraction

Data were extracted using a pre-designed extraction tool and all included studies were quality appraised using the CASP quality appraisal tools. Review authors (KO and JB) independently assessed the quality and data extraction for all potential included studies identified through the search strategy, using the selection criteria. This was conducted by checking for accuracy and consensus was reached via discussion. If it was unclear from the title or abstract whether a study met the criteria, or there was a disagreement over eligibility, the study was independently arbitrated by another author (JS), independently. The quality of all included studies was assessed using appropriate appraisal checklist developed by the Critical Appraisal Skill Programme (CASP). [18] This checklist contains twelve criteria which focus on the research focused issues, recruitment, accurately measured risk to minimised bias

(See Risk of bias interpretation Table 1), confounding factors, follow up, precision and trustworthiness of the results, and implication of study for practice and limitations. A pre-designed data extraction tool was employed to aid consistency with data extraction to include key features of the studies including setting, design, participants, method of data collection, results, discussion, and limitations.

### Risk of bias

Risk of bias interpretation is included in table 1.

### Patient and public involvement

No patient involved.

## Results

An initial search identified 70 studies (Figure 1). Nine studies were assessed as being suitable for full review. Following full review four studies were eligible for full critical appraisal and were therefore included for further analysis. Among these, two were based in Australia, one in England and one in Northern Ireland (Table 2).

### Risk of bias

A pre-designed data extraction tool was employed to aid consistency with data extraction to include key features of the studies including setting, design, participants, method of data collection, results, discussion, and limitations. All four cohort studies were subjected to quality assessment via risk of bias evaluation, this is included in table 1. All included studies had a low or very low risk of bias and all studies focused on long lie falls as an outcome. Only one study (Simpson et al. [15]) was considered to have a possible source of bias relating to statistical variables, although this was considered to be minimal.

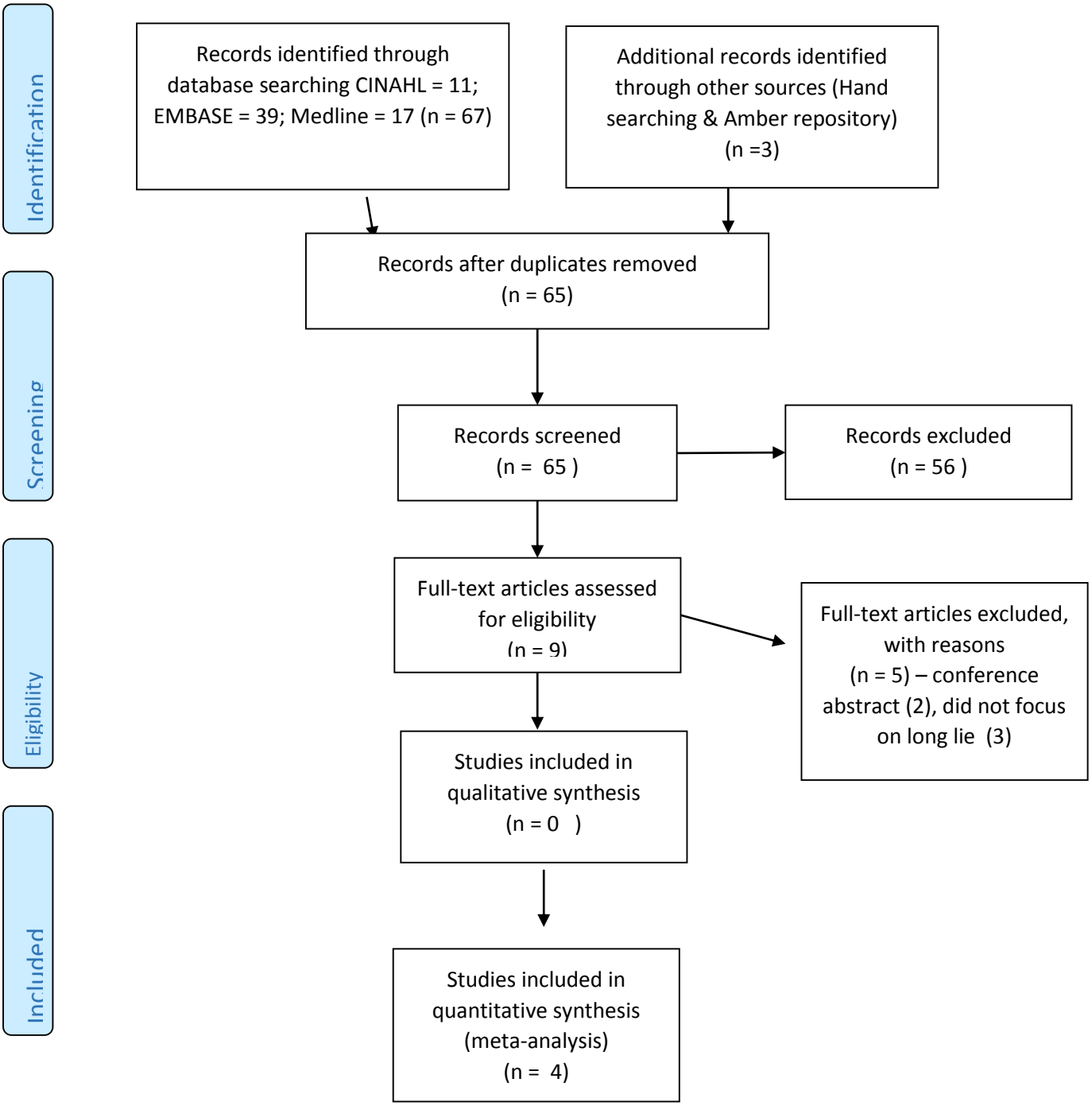
| Study | Potential source of bias | Study parameters | Assessed risk of bias |
|---|---|---|---|
| Scott (2020) | Was selection of exposed cohorts drawn from the same population? | All individuals aged ≥ 65 years in the geographical area covered by the South Eastern division of the Northern Ireland Ambulance Service (NIAS) who were attended by an NIAS paramedic for a fall and referred to the falls prevention service during the 3-month period from 1 July to 30 September 2017 were included. | None |
| | Can we be confident in the assessment of exposure? | Participants were identified using NIAS Emergency Ambulance Control (EAC) call data for incidents coded by the Advanced Medical Priority Dispatch System (AMPDS) as 'falls' and/or those whose main complaint recorded on Patient Report Forms (PRFs) by the ambulance crew was a fall. Re-contacts were identified by searching the EAC database for subsequent contacts with the NIAS within 1-month and 6-month follow-up. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – the cohort looked at re-contacts with the ambulance service so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Outcomes were measured using PRF data and the EAC control database. Clinical data, including medical history, drug history, clinical observations, treatments and onset time of the fall, were collected from PRF data. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | Yes - outcomes were measured using PRF data and the EAC control database. Clinical data, including medical history, drug history, clinical observations, treatments and onset time of the fall, were collected from PRF data. | Very low |
| | Can we be confident in the assessment of outcome? | Yes - the primary outcome was the rate of subsequent contacts with the ambulance service following referral. There were 1079 incidents of falls in service users aged ≥ 65 years. A referral rate of 7% (n = 75) was reported, constituting the study cohort. Re-contact rates were 37.3% (n = 28) within 1 month and 70.7% (n = 53) within 6 months. | Very low |
| | Was the follow-up of cohorts adequate? | Participants were followed up for 6 months. | Low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Fleming & Brayne (2008) | Was selection of exposed cohorts drawn from the same population? | Both groups comprised surviving participants drawn from the same cohort; the Cambridge City over-75s Cohort, a population based sample. | None |

| | | | |
|---|---|---|---|
| | Can we be confident in the assessment of exposure? | Falls data was obtained when reported by telephone, via a weekly falls report or follow-up visit or phone call by study nurse. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Yes - odds ratios and hazard ratios were adjusted for age, sex, and mobility. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | The longitudinal cohort study of older old people initially recruited participants through general practices in the 1980s, when they were all aged 75 or over. Repeated surveys since baseline have gathered data on a range of variables including socio-demographics, physical and mental health, function, and detailed cognitive assessment that included the mini-mental state examination. | Very low |
| | Can we be confident in the assessment of outcome? | Data recorded after each fall included whether the individual who fell had been able to get up without help, how long they were on the floor, any injuries, and whether they called for assistance. | Very low |
| | Was the follow-up of cohorts adequate? | Participants were followed-up intensively for 1 year. | Very low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Johnston et al. (2010) | Was selection of exposed cohorts drawn from the same population? | All data was obtained from database files provided by SAAS (South Australian Ambulance Service). | None |
| | Can we be confident in the assessment of exposure? | Excluding alarm activation error, all patients ≥ 65 years of age who used a personal alarm to call the ambulance were identified. | Low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – the primary outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Cases were matched by date of service, ambulance action (i.e. transport to hospital or lift only) and age with patients who did not use an alarm. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | A purpose-built data extraction file was used to record de-identified information extracted from each falls case including demographics: age, gender service details: date, type of service, response time fall details: mechanism, location, other persons present, location of person when | Very low |

| | | | |
|---|---|---|---|
| | | ambulance arrived, reported injury from fall ambulance contact details: alarm use, who called ambulance fall risk factors: medical conditions, medications, cognitive status, balance/mobility issues, continence, falls history, visual/sensory problems, environmental factors. | |
| | Can we be confident in the assessment of outcome? | The study outcomes were falls, or alert button activation error. These were identified by codes allocated when individuals activated a personal alarm, an ambulance was dispatched, but at the time of ambulance attendance the individual did not require either a lift, or transport to hospital. | Low |
| | Was the follow-up of cohorts adequate? | Participants were followed up for 1 month. | Low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Simpson et al. (2014) | Was selection of exposed cohorts drawn from the same population? | This study was a retrospective descriptive analysis of routinely collected data from the Ambulance Service of New South Wales (ASNSW), Australia.<br>Both cohorts were drawn from databases of the Ambulance Service of New South Wales (ASNSW). | None |
| | Can we be confident in the assessment of exposure? | Exposure is whether or not ambulance transport was required. Data contained dispatch and 'triple zero' call information from the Medical Priority Dispatch System (MPDS) software. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | A series of unadjusted analyses were conducted considering prognostic variables on a one-at-a-time basis. | Possible |
| | Can we be confident in the assessment of the presence of prognostic factors? | Data contained demographic and clinical information about the patient, including basic physiological measurements, identified injuries, and paramedic treatment. | Very low |
| | Can we be confident in the assessment of outcome? | MPDS includes a falls category. | Low |
| | Was the follow-up of cohorts adequate? | Participants were followed-up for 1 year. | Very low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |

*Table 1: Risk of bias of studies*

Figure 1: PRISMA Flow Diagram



## Study characteristics

Three of the four studies reported on people aged 65 years and older [15 19 20] with one study reporting on people aged over 90 years. [14] The included studies utilised different methodological approaches to examining long lie in the elderly including a prospective observational design [15] prospective cohort study [14] a retrospective audit [19] and a retrospective cross-sectional cohort study. [20]

## Determinants of a long lie

Whilst all studies focused on long lie as an outcome, there were some differences in the measures used to examine the experience of a long lie. Fleming and Brayne [14] focused on participants inability to get up without help, with a specific focus on the availability and use of call alarm systems. Johnston et al. [19] also looked at personal alarm use by examining the differences in fall characteristics and emergency service response to older fallers considering their use, or not, of a personal alarm. Simpson [15] focused on patient characteristics of those people confirmed to have fallen by paramedics arriving at the scene. Scott [20] focused on post-fall contacts of people referred to a falls prevention service with a specific focus on the outcomes of those with cognitive impairment and exposed to long lies.

## Ambulance response and patient outcomes

A prospective observational study, in New South Wales, Australia, including people aged 65 years and older who received an emergency response and were confirmed to have fallen by paramedics at the scene was reported by Simpson. [15] Fall-specific information, ambulance clinical records and computer-aided dispatch data was collected, with an emergency call system ambulance response defined as priority of 1 (urgent) or 2 (non-urgent). Those incidents allocated a response priority of 3–9 (cases not originating via the emergency call system including routine transports, operational standbys, medical appointments, and inter hospital transfers) were excluded. Over a 9-month period, 1,720 cases were submitted with the final sample comprising of 1,610 patient cases with a mean age of 83(8) and 61% of patients being female. The results demonstrated a significant relationship between increased age and suffering a fall ($p = 0.002$) with 44% (n=712) of fallers never having previously called an ambulance due to a fall, whilst 15% (n=248) had required an ambulance in the past month. The majority of falls occurred in a community dwelling (n=1018) with the most common location being the bedroom (23%) and living/dining room (20%). Eighty percent of falls (1,274/1,610) were from a standing height, followed by falls from bed and on stairs (each 7%) with the most commonly reported reason for falling being “loss of balance” (30%) or a “trip” (25%). In just over half of the cases the patient was still on

the floor when the ambulance arrived (53%) with 13% of these patients experiencing a long lie. The authors concluded trips decreased and loss of balance increased as age group increased.

Patient outcomes were also the focus of Scott [20] who focused on the characteristics, specifically cognitive impairment and long lie, and epidemiology of patients who re-contact the ambulance service in the subsequent 6-months following referral to a falls prevention service due to an ambulance response for a fall. Using a retrospective cross-sectional cohort design, the study identified 1079 calls to the ambulance service for people aged ≥ 65 years and over for a fall, with 7% (n=75) of these fallers resulting in a referral to a falls prevention service. A total of 136 re-contacts were made, with data on reasons for repeat contacts being available for 101 of these individuals. Repeat falls were responsible for 58% (n=59) of re-contacts. Long lies occurred in 81.2% (n = 56) of cases and 78.2% (n = 18) of patients exposed to a long lie re-contacted the service at least once within 6 months. Twenty patients re-contacted due to cognitive impairment.

Fleming and Brayne [14] reported data following a 1 year prospective cohort study of ageing, using fall calendars, phone calls, and visits in the community and care homes in in one city in England. Participants included 90 women and 20 men aged over 90 (n=110), who were the surviving participants of the Cambridge City over-75s Cohort. The study focused on participants inability to get up without help, lying on floor for a long time after falling, associated factors; availability and use of call alarm systems; participants'views on using call alarms to summon help if needed after falling. In this study, the authors found that call alarm systems were widely available but were often not used and almost all patients of those who lay on the floor for a long time and had a call alarm system did not activate it (97%). Lying on the floor for over an hour was strongly associated with serious injuries, admissions to hospital, and subsequent moves into long term care.

## Presence and impact of alarm use

Johnston et al. [19] examined the differences in fall characteristics and emergency service response to older fallers (≥ 65 years of age), considering their use, or not, of a personal alarm using a retrospective one-month audit of South Australian Ambulance Service (SAAS) records. Data on two samples of fallers were obtained consisting of cases between the specified study period coded as "falls" or "alarm activations" or "alert button activation error" (sample 1, n=1700) and a second sample which excluded alarm activation errors (n=224) but included those people ≥ 65 years of age who used a personal alarm to call the ambulance (n=124), in order to determine the scope of ambulance call outs for all falls and

alarm activations. The data demonstrated that in Sample 1, 379 of 1700 callers used a personal alarm to call the ambulance, with 58% of these alarm-activated calls being false alarms (neither lift nor hospital transport service provided). In Sample 2 the majority of alarm calls were made by females (72%) who were alone at the time of fall (78%). Long lies (> one hour) were reported in 11 cases of non-alarm users and a further 13 cases had an unspecified length of time on the floor following a fall before the ambulance arrived. The authors concluded that personal alarms should not be considered a reliable preventative measure for avoiding a long lie.

In their community based research, Fleming and Brayne [14] identified that call alarm systems were widely available but were often not used by participants, despite their data revealing virtually no one who did use an alarm lay more than an hour on the floor, yet nearly all (97%) of those who lay on the floor for a long time and had a call alarm system did not activate it.

### Prevalence of long lie

The studies included in this review identified that long lies were relatively uncommon and found limited evidence on the physical and clinical impacts of older people experiencing a long lie and requiring an ambulance response. In a sample of 110 patients, Fleming and Brayne [14] found that laying on the floor for over an hour was strongly associated with serious injuries, admissions to hospital, and subsequent moves into long term care. In a sample of patients who did or did not use their personal alarm to call the ambulance (n=124), Johnston et al. [19] found long lies were reported in 11 cases of non-alarm users and a further 13 cases had an unspecified length of time on the floor following a fall before the ambulance arrived. Scott [19] found that long lies were documented for 56 out of a total of 75 service users requiring an ambulance following a fall and 41.1% of these patients experienced a long lie. The study by Simpson [15] found that from a sample of 1610 patient cases, 13% of the 53% of patients who were still on the floor when the ambulance arrived experienced a long lie.

## Discussion

This review synthesised evidence exploring the relationship between a non-traumatic fall in the older person resulting in a long lie and poorer physical and clinical health outcomes. Limited evidence was found surrounding the long term outcomes of long lies with only four cohort studies meeting the inclusion criteria. Two studies specifically reported the effect of long lies on health

outcomes. [14 19] The quality of all included studies was assessed using appropriate appraisal checklist developed by the Critical Appraisal Skill Programme (CASP). [18] All four cohort studies were subjected to quality assessment via risk of bias evaluation and had a low or very low risk of bias with all studies focusing on long lie as an outcome.

All studies reported that the majority of falls in the community occurred in the home setting, with the person predominantly being found alone when the ambulance service arrived. Only a small number of ambulance calls following a fall resulted in a long lie, attributed to the rapid response times of the ambulance service, which prevented the manifestation of fall-related complications. Transport to hospital for fall-related injuries were relatively uncommon, which is consistent with previous research. [11] All included studies suffered from the methodological limitation of a reliance on faller recall or clinical documentation to determine the amount of time the person had been on the floor prior to calling the ambulance, meaning it is difficult to establish the exact length of time between experiencing a fall and receiving assistance.

Whilst a significant proportion of calls to the ambulance do not result in traumatic injury to the patient, the evidence suggests that there is a relationship between falls in the community setting and increased future morbidity, with many people necessitating future fall-related healthcare assistance and support. Indeed, Flemming and Brayne [14] found that serious injury was consistently associated with lying on the floor above one hour with 60% of those who were found on the floor following ambulance assistance subsequently admitted to hospital with a fall-related issue. Furthermore, in this study, long lies were associated with subsequent falls and future moves into long-term care demonstrating the potential long-term future health implications that experiencing a long lie may entail. Scott [20] also found that patients experiencing a long lie were also more likely to require future fall-related ambulance assistance.

Although not a focus of this review, the finding that personal alarm use was minimal in two studies is an interesting area of exploration. The reluctance to activate an alarm call from fallers found by Flemming and Brayne [14] could be consistent with evidence demonstrating that the non-use of alarms is associated with being unable to access the alarm, perceived patient burden and potential consequences of alarm activation. [19] Although alarm use should not necessarily be considered a preventative measure for falling or long lie [14] some evidence points towards their value in supporting people who experience a fall and require help. [21] It appears that the hesitancy of alarm use is concomitant with fear and anxiety around their usage and suggests that there is a need for teaching and support for the older patient in activating the alarm and supporting older people through

education on the effective and appropriate use of personal alarm systems could prove valuable in limiting the effects of experiencing a long lie. Newer technologies to alleviate the need for falls victims to personally trigger alarms, or the potential for big dataset interrogations to better describe health outcomes.

Future complications as a result of experiencing a long lie is not a new phenomenon; Wild et al. [22] found that older people experiencing a long lie were five times more likely to die within one year than age and sex matched controls, indicative of the vulnerability a long lie can have on ill-health. In a community based cohort study examining the predictors and prognosis associated with the inability to get up following a fall in people aged 72 years and over, Tinetti et al. [23] found that being unable to get up was associated with an increased risk of future hospitalisation, reduced mobility and an increased risk of mortality than those fallers who could get up, although these findings were not significant. Although there appears to be a relationship between long lies and prospective poorer health outcomes, this still remains a poorly understood area of investigation. Reducing long-lies is an important aspect of healthcare delivery with the potential to limit avoidable harms and the associated morbidity and mortality concomitant with falling in the older person. However, the most prominent finding in this research is the lack of available evidence surrounding the relationship between a non-traumatic fall in the older person resulting in a long lie and poorer physical and clinical health outcomes. This is partly due to a lack of robust evidence demonstrating the ill-effects of a long lie but also due to difficulties capturing details pertaining long lie such as the length of time between a fall and a call for help, which can be difficult to obtain through patient recall. This finding points to the need for the exploration into newer technologies to overcome the need for the individual to personally trigger fall alarms, as well as qualitative research to examine the lived experience of fall technology and experiencing a fall resulting in a long lie. The emotional consequences associated with falling in older adults should also be acknowledged, and a fear of falling or experiencing a fall in the absence of traumatic injury, can result in reduced self-confidence and a reluctance to engage in everyday activities [24] and can place increased demands on caregivers for support. [25] Exploration and analysis of big datasets to better understand and identify health outcomes of fall victims could also increase understanding of the future complications and long-term outcomes of experiencing a long lie and contribute towards the currently limited evidence base in this important area of research.

Given the increasing number of older people living alone or with the support from carers, examining the lived experience of a long lie, potential future complications and views and experiences of using personal alarms, could provide valuable insights in how to support this vulnerable patient group.

Future research should also focus on evolving interventions for wearable, smart and e-technology for automatic fall detection. Indeed, Wang et al. [26] provide a comprehensive review of fall detection systems which are designed to either detect or predict a fall and encompass a range of wearable, visual, ambient and sensor fusion systems. Whilst the authors state a combination of visual and wearable devices offer the most accurate method of fall detection, additional research is required for accurate benchmarking between different interventions. Such research may overcome some of the limitations associated with personal alarm use and enable strategies for reducing the potential negative consequences of suffering a fall in the community setting with an associated long lie.

## Limitations

Experiencing a long lie fall in the older person is an important area of concern in healthcare, however a reliance on patient or carer recall to determine the length of a long lie means that obtaining an accurate measure of the physical and clinical health outcomes of long lie in the older person is difficult to quantify. These inconsistencies can lead to inaccurate reporting and measuring of the long-term implications of a long lie fall. Moreover, the papers included in this review had an overreliance on the use of personal alarms in preventing long lies in the older person, rather than a specific focus on the long-term outcomes associated with a long lie fall, perhaps attributable to the lack of robust clinical evidence. Thus, the physical implications of being unable to get up from the floor following a non-traumatic fall and the potential threat of pressure related damage or injury is difficult to understand. This is important, as the potential of developing a pressure related injury due to limited mobility is greatly increased in the older person population. Although all the studies reported in this review defined a long lie as being unable to get up from the floor for <60 minutes following a fall, a lack of a formal definition of a long lie means that some studies could have been excluded from this review. Only 4 cohort studies were included in this review therefore there could be a potential for risk of bias in the findings. However, utilising risk of bias interpretation tool for cohort studies, this review reported a low risk of bias and this points to the need for more research into the experiences of a long lie fall in the older person. Future research should focus on a consensus for defining long lie fall and developing robust methodologies for accurately determining the physical and clinical impacts of long lie in the older person.

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| Author/year/ Country | Design/setting/ population details | Outcomes being assessed | Main findings | Conclusion |
|---|---|---|---|---|
| Johnston, K., Worley, A., Grimmer-Somers, K., Sutherland, M., & Amos, L. (2010). Australia | A retrospective one month audit of South Australian Ambulance Service records (n=1700) | Characteristics of ambulance call outs for falls<br>Use of personal alarms by the fallers | Ambulance response time did not differ between alarm and non-alarm users.<br>82% of alarm users and 75% of non-alarm users were on the ground when the ambulance arrived.<br>11 of non-alarm users self-reported 'long lies' (>one hour) before the ambulance was called<br>There were 13 other cases with unknown time on the floor. | Older women living alone were the major users of personal alarms for assistance after falling.<br>If activated quickly, alarms enabled most fallers to gain ambulance attention within 15 minutes.<br>Personal alarm use was associated with a high incidence of false alarms. |
| Fleming, J., & Brayne, C. (2008). England | 1 year follow-up of participants in a prospective cohort study of ageing, using fall calendars, phone calls, and visits (n=110) | Inability to get up without help, lying on floor for a long time after falling, associated factors; availability and use of call alarm systems.<br>Participants' views on using call alarms to summon help if needed after falling. | 54% of fall reports described the participant as being found on the floor and 82% of falls occurred when the person was alone.<br>Of the 60% who fell, 30% had lain on the floor for an hour or more.<br>Difficulty in getting up was consistently associated with age, reported mobility, and severe cognitive impairment.<br>Cognition was the only characteristic that predicted lying on the floor for a long time. | This study identifies the oldest people are more at risk of falls and that the design of personal alarms requires development. Call alarms were widely available but were not used in most cases of falls. |

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| | | | Lying on the floor for a long time was strongly associated with serious injuries, admission to hospital, and subsequent moves into long term care. | |
| Simpson, P. M., Bendall, J. C., Tiedemann, A., Lord, S. R., & Close, J. C. T. (2014). Australia | Prospective cohort study of people aged ≥65 who had fallen and called for an ambulance (n=1610) | Collection of fall-related data allowing for a sequential description of fall-related cases from time of ambulance dispatch through to the end of the prehospital episode of care. | From 1,610 eligible cases and "long-lies" (>60 minutes on the ground) occurred in 13% of cases. Patients were predominantly female and community dwelling.<br><br>The most common patient-reported reasons for falling were:<br>Loss of balance<br>Simple trips<br>New injury and/or pain was documented for 1,172 of patients<br>656 presented with abnormal" physiology<br>Only 238 presented with no new injury/pain and normal physiology. | The paper concludes ambulance services appear to provide timely responses to older people who have fallen, and "long-lies" are relatively uncommon. |
| Scott, J (2020) | Retrospective cross-sectional cohort study in South Eastern division of the Northern Ireland Ambulance Service (NIAS). | Subsequent contacts with the ambulance service following referral. | There were 1079 incidents of falls in service users aged ≥ 65 years with a referral rate of 7% (n = 75) was reported. Re-contact rates were 37.3% (n = 28) within 1 month and 70.7% (n = 53) within 6 months.<br><br>Women and those experiencing a 'long lie' were more likely to re- | Interventions targeted at reducing long lies and supporting patients with cognitive impairments should be explored. Clinical documentation needs to be improved. |

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|---|---|---|---|---|
| | | | contact. Those with cognitive impairment were vulnerable to falls and repeat falls. Documentation for recording patient-related information was poor. | |

*Table 2: Table of included studies*

# Statements

## Contributorship

JB was the primary author. All authors made substantial contributions to the design, analysis and interpretation of this review. All authors were involved in drafting of the work and the development and review and revision of the final manuscript.

## Registration and protocol.

This review was not registered and a protocol was not prepared.


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## Competing interests

There are no competing interests.

## Data Sharing/Data availability

This manuscript does not use any patient identifiable information.

## Ethics Approval

Ethics Approval - Not Applicable

## PPI

No patients were involved

Manuscript with changes highlighted/marked


## Abstract

**Background:** The long term impacts of experiencing a 'long lie' following a fall in the older person are poorly understood. This systematic review explored the impact of a long lie fall on physical and clinical outcomes in older people requiring an ambulance.

**Methods:** PRISMA guidelines were followed.

**Results:** 70 studies were identified. Nine studies were suitable for full review. Four studies meeting the inclusion criteria were included. The Critical Appraisal Skill Programme (CASP) assessed the quality of all included studies. Three studies reported on people aged 65 years and older. One study reported on people aged over 90 years. Personal alarm use was examined in two studies. One study explored patient characteristics of people confirmed to have fallen by paramedics at the scene. One study examined re-contact and characteristics of fallers referred to a falls prevention service.

**Discussion:** Cognitive impairment and long lie were a caveat for falls and repeated falls. Personal alarm use was infrequent suggesting a need for supporting the older patient in appropriate alarm use and exploration of newer technologies to alleviate their need. Future research should focus on interventions for wearable, smart and e-technology for automatic fall detection and qualitative exploration of the lived experience.



## Key Words

Long lie; Falls; Elderly; Ambulance response; Fall detection technology; Lived experience; Patient experience

# Introduction

Falls are a common occurrence and are the largest cause of hospital admission in the older person; between 2017-2018 approximately 220,160 emergency hospital admissions in the United Kingdom (UK) were attributable to falls in this patient group (aged 65 years and over) with just over 66% of these patients aged over 80 years. [1] Hospital-related falls account for approximately 240,000 reported in-patient safety incidents in UK hospitals [2], with an estimated cost of 2.3 billion. [3] A fall is defined as an event which causes a person to, unintentionally, rest on the ground or other lower level. For the purposes of this paper, a non-traumatic fall is defined as a fall not causing significant trauma or injury to the patient. [4]

The prevention and management of falls in older people have been the focus of UK national government initiatives which have endeavoured to focus on the reduction of morbidity and mortality in this patient group. [5, 6] The ambulance service is under growing pressure to provide care provision to an increasing number of older people who fall in their own home [7]; a frequent reason for an emergency ambulance call. [8] Even in those people who do not necessitate hospital admission, there is an increased likelihood of poorer health outcomes, an increased risk of future admission to hospital and increased morbidity than the general older person population. [9] Community related falls are largely preventable, however, data suggests that approximately 1 in 3 adults over 65 will experience one fall annually [3] and an estimated 5% of falls in the community result in hospitalisation for injuries such as fractures. [10] Snooks et al [11] found that 8% of all calls made to the London Ambulance Service were due to older people who had fallen at home, with 36% of these people not requiring an admission to hospital, despite 49% of fallers requiring healthcare assistance in the two-weeks post-fall, with an increased risk of death and hospital admission than the general population.

Clinical and physical age-related complications such as muscle weakness, visual impairment or loss of balance, as well as environmental factors such as a hazardous environment increase fall-related risk. Hazards in the home resulting in a fall cost the NHS in the UK approximately £435 million [12] and adapting the home environment to improve safety and limit fall-risk has been a focus of interventions in falls prevention. [13] Suffering a fall at home can be a precursor to further future falls and the rehabilitative resources required to manage an older person post-fall can place significant demands on healthcare services. [3]

**Experiencing a long lie fall**

A potentially serious complication of community-related falls in older people is the inability to get up following a fall, which is associated with severe psychological, emotional and physical distress connected to a fear of future falls, pressure-related injuries and loss of mobility. [3] Providing information on how to avoid a 'long lie', generally defined as being unable to get up from the floor for over one hour following a fall, is a priority in the NICE recommendations for preventing falls in older people. [3] However, there is scant evidence surrounding the long-term outcomes of people who experience a long lie fall in the community setting. Fleming et al [14] undertook a prospective cohort study in the UK to examine the use of call alarm systems and their association with experiencing a long lie in people over 90 years of age who had fallen at home. In a sample of 110 participants, at one-year follow-up 60% were reported to have experienced a fall on at least one occasion and falls in the community setting accounted for 45% of all reported falls. Interestingly, just over half of all reported falls resulted in the participant being found on the floor, with the majority of these participants requiring assistance to get up from the floor. A total of 15% of all reported falls resulted in a long lie, however, accounting for the individual people who fell rather than the reported number of falls, long lies represented 30% of falls. Those who did experience a long lie were more likely to suffer subsequent serious injuries, admissions to hospital and future long term care. Conversely, Simpson et al [15] explored the characteristics of older people who had fallen in Australia and called an ambulance through a prospective cohort study from a population of 1,610 cases and found that just over 53% of patients were still on the floor when the ambulance arrived. Due to the quick ambulance response time, only 13% of these patients experienced a "long-lie". Cannon et al [16] found no relationship between ambulance response time and long lie in a sample of 503 patients.

The physical and clinical impacts on the older person experiencing a long lie remain poorly understood with variations in the patient population investigated, study methodologies and a lack of robust clinical evidence contributing to a lack of understanding around the long term impacts of a long lie fall. As the ambulance service are required to respond to an increasing number of older people who fall, understanding the physical and clinical impacts of a fall is fundamental to limiting the number of people suffering a long lie.

## Aims and objectives

This systematic review aimed to explore the relationship between a non-traumatic fall in people aged 65 years or over resulting in a long lie (defined as remaining on the floor for an hour or longer) and subsequently cared for by the ambulance service, and the associated physical and clinical health related outcomes. The following objectives were defined to achieve this aim:

1. To determine if there is an association between length of time spent on the floor after falling and physical and clinical outcomes for older patients.

2. To understand whether the type and timeliness of response provided by the ambulance service has any physical or clinical consequences for older patients who have a long-lie after falling.

# Methods

The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) guidelines were followed (Figure 1). [17] Adapted critical appraisal tools [18] were used to assess the papers.

## Inclusion and exclusion criteria

The population of interest were patients over 65 years of age who suffered a non-traumatic fall and subsequent long lie (defined as remaining on the floor for >60 minutes), and who were cared for by the ambulance service. Randomised controlled trials (RCTs), cohort, observational studies whether prospective or retrospective in design, and audits were suitable for inclusion. No constraints were placed on language of publication or date of publication. Conference papers, poster abstracts, opinion papers and case studies were excluded.

## Search strategy.

PubMed, Embase, Medline, CINAHL, and the Amber repository were searched from 1980 to March 2021 using keywords and medical subject headings (mesh) determined by the authors: ("EMERGENCY MEDICAL SERVICES"/ OR AMBULANCES/ OR (paramedics).ti,ab) AND ((long).ti,ab AND (lie OR lay). The study team also reviewed the reference lists of retrieved studies to identify studies, which had not been identified by the search strategy. Duplicate studies were excluded.

## Quality assessment and data extraction

Data were extracted using a pre-designed extraction tool and all included studies were quality appraised using the CASP quality appraisal tools. Review authors (KO and JB) independently assessed the quality and data extraction for all potential included studies identified through the search strategy, using the selection criteria. This was conducted by checking for accuracy and consensus was reached via discussion. If it was unclear from the title or abstract whether a study met the criteria, or there was a disagreement over eligibility, the study was independently arbitrated by another author (JS), independently. The quality of all included studies was assessed using appropriate appraisal checklist developed by the Critical Appraisal Skill Programme (CASP). [18] This checklist contains twelve criteria which focus on the research focused issues, recruitment, accurately measured risk to minimised bias

(See Risk of bias interpretation Table 1), confounding factors, follow up, precision and trustworthiness of the results, and implication of study for practice and limitations. A pre-designed data extraction tool was employed to aid consistency with data extraction to include key features of the studies including setting, design, participants, method of data collection, results, discussion, and limitations.

### Risk of bias

Risk of bias interpretation is included in table 1.

### Patient and public involvement

No patient involved.

## Results

An initial search identified 70 studies (Figure 1). Nine studies were assessed as being suitable for full review. Following full review four studies were eligible for full critical appraisal and were therefore included for further analysis. Among these, two were based in Australia, one in England and one in Northern Ireland (Table 2).

### Risk of bias

A pre-designed data extraction tool was employed to aid consistency with data extraction to include key features of the studies including setting, design, participants, method of data collection, results, discussion, and limitations. All four cohort studies were subjected to quality assessment via risk of bias evaluation, this is included in table 1. All included studies had a low or very low risk of bias and all studies focused on long lie falls as an outcome. Only one study (Simpson et al. [15]) was considered to have a possible source of bias relating to statistical variables, although this was considered to be minimal.

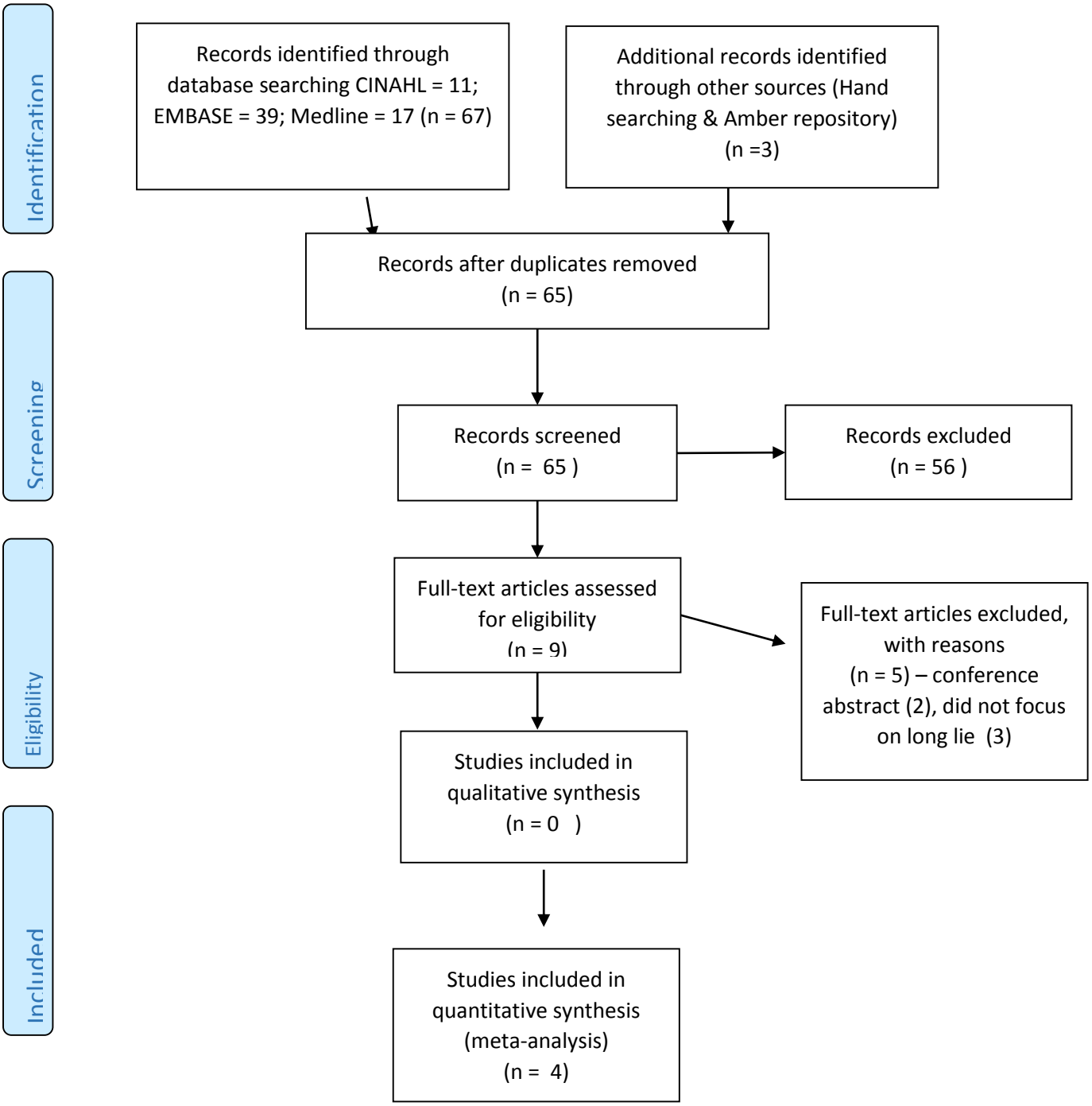
| Study | Potential source of bias | Study parameters | Assessed risk of bias |
|---|---|---|---|
| Scott (2020) | Was selection of exposed cohorts drawn from the same population? | All individuals aged ≥ 65 years in the geographical area covered by the South Eastern division of the Northern Ireland Ambulance Service (NIAS) who were attended by an NIAS paramedic for a fall and referred to the falls prevention service during the 3-month period from 1 July to 30 September 2017 were included. | None |
| | Can we be confident in the assessment of exposure? | Participants were identified using NIAS Emergency Ambulance Control (EAC) call data for incidents coded by the Advanced Medical Priority Dispatch System (AMPDS) as 'falls' and/or those whose main complaint recorded on Patient Report Forms (PRFs) by the ambulance crew was a fall. Re-contacts were identified by searching the EAC database for subsequent contacts with the NIAS within 1-month and 6-month follow-up. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – the cohort looked at re-contacts with the ambulance service so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Outcomes were measured using PRF data and the EAC control database. Clinical data, including medical history, drug history, clinical observations, treatments and onset time of the fall, were collected from PRF data. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | Yes - outcomes were measured using PRF data and the EAC control database. Clinical data, including medical history, drug history, clinical observations, treatments and onset time of the fall, were collected from PRF data. | Very low |
| | Can we be confident in the assessment of outcome? | Yes - the primary outcome was the rate of subsequent contacts with the ambulance service following referral. There were 1079 incidents of falls in service users aged ≥ 65 years. A referral rate of 7% (n = 75) was reported, constituting the study cohort. Re-contact rates were 37.3% (n = 28) within 1 month and 70.7% (n = 53) within 6 months. | Very low |
| | Was the follow-up of cohorts adequate? | Participants were followed up for 6 months. | Low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Fleming & Brayne (2008) | Was selection of exposed cohorts drawn from the same population? | Both groups comprised surviving participants drawn from the same cohort; the Cambridge City over-75s Cohort, a population based sample. | None |

| | | | |
|---|---|---|---|
| | Can we be confident in the assessment of exposure? | Falls data was obtained when reported by telephone, via a weekly falls report or follow-up visit or phone call by study nurse. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Yes - odds ratios and hazard ratios were adjusted for age, sex, and mobility. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | The longitudinal cohort study of older old people initially recruited participants through general practices in the 1980s, when they were all aged 75 or over. Repeated surveys since baseline have gathered data on a range of variables including socio-demographics, physical and mental health, function, and detailed cognitive assessment that included the mini-mental state examination. | Very low |
| | Can we be confident in the assessment of outcome? | Data recorded after each fall included whether the individual who fell had been able to get up without help, how long they were on the floor, any injuries, and whether they called for assistance. | Very low |
| | Was the follow-up of cohorts adequate? | Participants were followed-up intensively for 1 year. | Very low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Johnston et al. (2010) | Was selection of exposed cohorts drawn from the same population? | All data was obtained from database files provided by SAAS (South Australian Ambulance Service). | None |
| | Can we be confident in the assessment of exposure? | Excluding alarm activation error, all patients ≥ 65 years of age who used a personal alarm to call the ambulance were identified. | Low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – the primary outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | Cases were matched by date of service, ambulance action (i.e. transport to hospital or lift only) and age with patients who did not use an alarm. | Very low |
| | Can we be confident in the assessment of the presence of prognostic factors? | A purpose-built data extraction file was used to record de-identified information extracted from each falls case including demographics: age, gender service details: date, type of service, response time fall details: mechanism, location, other persons present, location of person when | Very low |

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| | | ambulance arrived, reported injury from fall ambulance contact details: alarm use, who called ambulance fall risk factors: medical conditions, medications, cognitive status, balance/mobility issues, continence, falls history, visual/sensory problems, environmental factors. | |
| | Can we be confident in the assessment of outcome? | The study outcomes were falls, or alert button activation error. These were identified by codes allocated when individuals activated a personal alarm, an ambulance was dispatched, but at the time of ambulance attendance the individual did not require either a lift, or transport to hospital. | Low |
| | Was the follow-up of cohorts adequate? | Participants were followed up for 1 month. | Low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |
| Simpson et al. (2014) | Was selection of exposed cohorts drawn from the same population? | This study was a retrospective descriptive analysis of routinely collected data from the Ambulance Service of New South Wales (ASNSW), Australia.<br>Both cohorts were drawn from databases of the Ambulance Service of New South Wales (ASNSW). | None |
| | Can we be confident in the assessment of exposure? | Exposure is whether or not ambulance transport was required. Data contained dispatch and 'triple zero' call information from the Medical Priority Dispatch System (MPDS) software. | Very low |
| | Can we be confident that the outcome of interest was not present at the start of the study? | Yes – outcome was experiencing a long lie following a fall so this is not relevant. | None |
| | Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? | A series of unadjusted analyses were conducted considering prognostic variables on a one-at-a-time basis. | Possible |
| | Can we be confident in the assessment of the presence of prognostic factors? | Data contained demographic and clinical information about the patient, including basic physiological measurements, identified injuries, and paramedic treatment. | Very low |
| | Can we be confident in the assessment of outcome? | MPDS includes a falls category. | Low |
| | Was the follow-up of cohorts adequate? | Participants were followed-up for 1 year. | Very low |
| | Were co-interventions similar between groups? | There were no co-interventions. | None |

*Table 1: Risk of bias of studies*

Figure 1: PRISMA Flow Diagram



## Study characteristics

Three of the four studies reported on people aged 65 years and older [15 19 20] with one study reporting on people aged over 90 years. [14] The included studies utilised different methodological approaches to examining long lie in the elderly including a prospective observational design [15] prospective cohort study [14] a retrospective audit [19] and a retrospective cross-sectional cohort study. [20]

## Determinants of a long lie

Whilst all studies focused on long lie as an outcome, there were some differences in the measures used to examine the experience of a long lie. Fleming and Brayne [14] focused on participants inability to get up without help, with a specific focus on the availability and use of call alarm systems. Johnston et al. [19] also looked at personal alarm use by examining the differences in fall characteristics and emergency service response to older fallers considering their use, or not, of a personal alarm. Simpson [15] focused on patient characteristics of those people confirmed to have fallen by paramedics arriving at the scene. Scott [20] focused on post-fall contacts of people referred to a falls prevention service with a specific focus on the outcomes of those with cognitive impairment and exposed to long lies.

## Ambulance response and patient outcomes

A prospective observational study, in New South Wales, Australia, including people aged 65 years and older who received an emergency response and were confirmed to have fallen by paramedics at the scene was reported by Simpson. [15] Fall-specific information, ambulance clinical records and computer-aided dispatch data was collected, with an emergency call system ambulance response defined as priority of 1 (urgent) or 2 (non-urgent). Those incidents allocated a response priority of 3–9 (cases not originating via the emergency call system including routine transports, operational standbys, medical appointments, and inter hospital transfers) were excluded. Over a 9-month period, 1,720 cases were submitted with the final sample comprising of 1,610 patient cases with a mean age of 83(8) and 61% of patients being female. The results demonstrated a significant relationship between increased age and suffering a fall ($p = 0.002$) with 44% (n=712) of fallers never having previously called an ambulance due to a fall, whilst 15% (n=248) had required an ambulance in the past month. The majority of falls occurred in a community dwelling (n=1018) with the most common location being the bedroom (23%) and living/dining room (20%). Eighty percent of falls (1,274/1,610) were from a standing height, followed by falls from bed and on stairs (each 7%) with the most commonly reported reason for falling being “loss of balance” (30%) or a “trip” (25%). In just over half of the cases the patient was still on

the floor when the ambulance arrived (53%) with 13% of these patients experiencing a long lie. The authors concluded trips decreased and loss of balance increased as age group increased.

Patient outcomes were also the focus of Scott [20] who focused on the characteristics, specifically cognitive impairment and long lie, and epidemiology of patients who re-contact the ambulance service in the subsequent 6-months following referral to a falls prevention service due to an ambulance response for a fall. Using a retrospective cross-sectional cohort design, the study identified 1079 calls to the ambulance service for people aged ≥ 65 years and over for a fall, with 7% (n=75) of these fallers resulting in a referral to a falls prevention service. A total of 136 re-contacts were made, with data on reasons for repeat contacts being available for 101 of these individuals. Repeat falls were responsible for 58% (n=59) of re-contacts. Long lies occurred in 81.2% (n = 56) of cases and 78.2% (n = 18) of patients exposed to a long lie re-contacted the service at least once within 6 months. Twenty patients re-contacted due to cognitive impairment.

Fleming and Brayne [14] reported data following a 1 year prospective cohort study of ageing, using fall calendars, phone calls, and visits in the community and care homes in in one city in England. Participants included 90 women and 20 men aged over 90 (n=110), who were the surviving participants of the Cambridge City over-75s Cohort. The study focused on participants inability to get up without help, lying on floor for a long time after falling, associated factors; availability and use of call alarm systems; participants'views on using call alarms to summon help if needed after falling. In this study, the authors found that call alarm systems were widely available but were often not used and almost all patients of those who lay on the floor for a long time and had a call alarm system did not activate it (97%). Lying on the floor for over an hour was strongly associated with serious injuries, admissions to hospital, and subsequent moves into long term care.

## Presence and impact of alarm use

Johnston et al. [19] examined the differences in fall characteristics and emergency service response to older fallers (≥ 65 years of age), considering their use, or not, of a personal alarm using a retrospective one-month audit of South Australian Ambulance Service (SAAS) records. Data on two samples of fallers were obtained consisting of cases between the specified study period coded as "falls" or "alarm activations" or "alert button activation error" (sample 1, n=1700) and a second sample which excluded alarm activation errors (n=224) but included those people ≥ 65 years of age who used a personal alarm to call the ambulance (n=124), in order to determine the scope of ambulance call outs for all falls and

alarm activations. The data demonstrated that in Sample 1, 379 of 1700 callers used a personal alarm to call the ambulance, with 58% of these alarm-activated calls being false alarms (neither lift nor hospital transport service provided). In Sample 2 the majority of alarm calls were made by females (72%) who were alone at the time of fall (78%). Long lies (> one hour) were reported in 11 cases of non-alarm users and a further 13 cases had an unspecified length of time on the floor following a fall before the ambulance arrived. The authors concluded that personal alarms should not be considered a reliable preventative measure for avoiding a long lie.

In their community based research, Fleming and Brayne [14] identified that call alarm systems were widely available but were often not used by participants, despite their data revealing virtually no one who did use an alarm lay more than an hour on the floor, yet nearly all (97%) of those who lay on the floor for a long time and had a call alarm system did not activate it.

### Prevalence of long lie

The studies included in this review identified that long lies were relatively uncommon and found limited evidence on the physical and clinical impacts of older people experiencing a long lie and requiring an ambulance response. In a sample of 110 patients, Fleming and Brayne [14] found that laying on the floor for over an hour was strongly associated with serious injuries, admissions to hospital, and subsequent moves into long term care. In a sample of patients who did or did not use their personal alarm to call the ambulance (n=124), Johnston et al. [19] found long lies were reported in 11 cases of non-alarm users and a further 13 cases had an unspecified length of time on the floor following a fall before the ambulance arrived. Scott [19] found that long lies were documented for 56 out of a total of 75 service users requiring an ambulance following a fall and 41.1% of these patients experienced a long lie. The study by Simpson [15] found that from a sample of 1610 patient cases, 13% of the 53% of patients who were still on the floor when the ambulance arrived experienced a long lie.

## Discussion

This review synthesised evidence exploring the relationship between a non-traumatic fall in the older person resulting in a long lie and poorer physical and clinical health outcomes. Limited evidence was found surrounding the long term outcomes of long lies with only four cohort studies meeting the inclusion criteria. Two studies specifically reported the effect of long lies on health

outcomes. [14 19] The quality of all included studies was assessed using appropriate appraisal checklist developed by the Critical Appraisal Skill Programme (CASP). [18] All four cohort studies were subjected to quality assessment via risk of bias evaluation and had a low or very low risk of bias with all studies focusing on long lie as an outcome.

All studies reported that the majority of falls in the community occurred in the home setting, with the person predominantly being found alone when the ambulance service arrived. Only a small number of ambulance calls following a fall resulted in a long lie, attributed to the rapid response times of the ambulance service, which prevented the manifestation of fall-related complications. Transport to hospital for fall-related injuries were relatively uncommon, which is consistent with previous research. [11] All included studies suffered from the methodological limitation of a reliance on faller recall or clinical documentation to determine the amount of time the person had been on the floor prior to calling the ambulance, meaning it is difficult to establish the exact length of time between experiencing a fall and receiving assistance.

Whilst a significant proportion of calls to the ambulance do not result in traumatic injury to the patient, the evidence suggests that there is a relationship between falls in the community setting and increased future morbidity, with many people necessitating future fall-related healthcare assistance and support. Indeed, Flemming and Brayne [14] found that serious injury was consistently associated with lying on the floor above one hour with 60% of those who were found on the floor following ambulance assistance subsequently admitted to hospital with a fall-related issue. Furthermore, in this study, long lies were associated with subsequent falls and future moves into long-term care demonstrating the potential long-term future health implications that experiencing a long lie may entail. Scott [20] also found that patients experiencing a long lie were also more likely to require future fall-related ambulance assistance.

Although not a focus of this review, the finding that personal alarm use was minimal in two studies is an interesting area of exploration. The reluctance to activate an alarm call from fallers found by Flemming and Brayne [14] could be consistent with evidence demonstrating that the non-use of alarms is associated with being unable to access the alarm, perceived patient burden and potential consequences of alarm activation. [19] Although alarm use should not necessarily be considered a preventative measure for falling or long lie [14] some evidence points towards their value in supporting people who experience a fall and require help. [21] It appears that the hesitancy of alarm use is concomitant with fear and anxiety around their usage and suggests that there is a need for teaching and support for the older patient in activating the alarm and supporting older people through

education on the effective and appropriate use of personal alarm systems could prove valuable in limiting the effects of experiencing a long lie. Newer technologies to alleviate the need for falls victims to personally trigger alarms, or the potential for big dataset interrogations to better describe health outcomes.

Future complications as a result of experiencing a long lie is not a new phenomenon; Wild et al. [22] found that older people experiencing a long lie were five times more likely to die within one year than age and sex matched controls, indicative of the vulnerability a long lie can have on ill-health. In a community based cohort study examining the predictors and prognosis associated with the inability to get up following a fall in people aged 72 years and over, Tinetti et al. [23] found that being unable to get up was associated with an increased risk of future hospitalisation, reduced mobility and an increased risk of mortality than those fallers who could get up, although these findings were not significant. Although there appears to be a relationship between long lies and prospective poorer health outcomes, this still remains a poorly understood area of investigation. Reducing long-lies is an important aspect of healthcare delivery with the potential to limit avoidable harms and the associated morbidity and mortality concomitant with falling in the older person. However, the most prominent finding in this research is the lack of available evidence surrounding the relationship between a non-traumatic fall in the older person resulting in a long lie and poorer physical and clinical health outcomes. This is partly due to a lack of robust evidence demonstrating the ill-effects of a long lie but also due to difficulties capturing details pertaining long lie such as the length of time between a fall and a call for help, which can be difficult to obtain through patient recall. This finding points to the need for the exploration into newer technologies to overcome the need for the individual to personally trigger fall alarms, as well as qualitative research to examine the lived experience of fall technology and experiencing a fall resulting in a long lie. The emotional consequences associated with falling in older adults should also be acknowledged, and a fear of falling or experiencing a fall in the absence of traumatic injury, can result in reduced self-confidence and a reluctance to engage in everyday activities [24] and can place increased demands on caregivers for support. [25] Exploration and analysis of big datasets to better understand and identify health outcomes of fall victims could also increase understanding of the future complications and long-term outcomes of experiencing a long lie and contribute towards the currently limited evidence base in this important area of research.

Given the increasing number of older people living alone or with the support from carers, examining the lived experience of a long lie, potential future complications and views and experiences of using personal alarms, could provide valuable insights in how to support this vulnerable patient group.

Future research should also focus on evolving interventions for wearable, smart and e-technology for automatic fall detection. Indeed, Wang et al. [26] provide a comprehensive review of fall detection systems which are designed to either detect or predict a fall and encompass a range of wearable, visual, ambient and sensor fusion systems. Whilst the authors state a combination of visual and wearable devices offer the most accurate method of fall detection, additional research is required for accurate benchmarking between different interventions. Such research may overcome some of the limitations associated with personal alarm use and enable strategies for reducing the potential negative consequences of suffering a fall in the community setting with an associated long lie.

## Limitations

Experiencing a long lie fall in the older person is an important area of concern in healthcare, however a reliance on patient or carer recall to determine the length of a long lie means that obtaining an accurate measure of the physical and clinical health outcomes of long lie in the older person is difficult to quantify. These inconsistencies can lead to inaccurate reporting and measuring of the long-term implications of a long lie fall. Moreover, the papers included in this review had an overreliance on the use of personal alarms in preventing long lies in the older person, rather than a specific focus on the long-term outcomes associated with a long lie fall, perhaps attributable to the lack of robust clinical evidence. Thus, the physical implications of being unable to get up from the floor following a non-traumatic fall and the potential threat of pressure related damage or injury is difficult to understand. This is important, as the potential of developing a pressure related injury due to limited mobility is greatly increased in the older person population. Although all the studies reported in this review defined a long lie as being unable to get up from the floor for <60 minutes following a fall, a lack of a formal definition of a long lie means that some studies could have been excluded from this review. Only 4 cohort studies were included in this review therefore there could be a potential for risk of bias in the findings. However, utilising risk of bias interpretation tool for cohort studies, this review reported a low risk of bias and this points to the need for more research into the experiences of a long lie fall in the older person. Future research should focus on a consensus for defining long lie fall and developing robust methodologies for accurately determining the physical and clinical impacts of long lie in the older person.

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| Author/year/ Country | Design/setting/ population details | Outcomes being assessed | Main findings | Conclusion |
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| Johnston, K., Worley, A., Grimmer-Somers, K., Sutherland, M., & Amos, L. (2010). Australia | A retrospective one month audit of South Australian Ambulance Service records (n=1700) | Characteristics of ambulance call outs for falls<br>Use of personal alarms by the fallers | Ambulance response time did not differ between alarm and non-alarm users.<br>82% of alarm users and 75% of non-alarm users were on the ground when the ambulance arrived.<br>11 of non-alarm users self-reported 'long lies' (>one hour) before the ambulance was called<br>There were 13 other cases with unknown time on the floor. | Older women living alone were the major users of personal alarms for assistance after falling.<br>If activated quickly, alarms enabled most fallers to gain ambulance attention within 15 minutes.<br>Personal alarm use was associated with a high incidence of false alarms. |
| Fleming, J., & Brayne, C. (2008). England | 1 year follow-up of participants in a prospective cohort study of ageing, using fall calendars, phone calls, and visits (n=110) | Inability to get up without help, lying on floor for a long time after falling, associated factors; availability and use of call alarm systems.<br>Participants' views on using call alarms to summon help if needed after falling. | 54% of fall reports described the participant as being found on the floor and 82% of falls occurred when the person was alone.<br>Of the 60% who fell, 30% had lain on the floor for an hour or more.<br>Difficulty in getting up was consistently associated with age, reported mobility, and severe cognitive impairment.<br>Cognition was the only characteristic that predicted lying on the floor for a long time. | This study identifies the oldest people are more at risk of falls and that the design of personal alarms requires development. Call alarms were widely available but were not used in most cases of falls. |

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| | | | Lying on the floor for a long time was strongly associated with serious injuries, admission to hospital, and subsequent moves into long term care. | |
| Simpson, P. M., Bendall, J. C., Tiedemann, A., Lord, S. R., & Close, J. C. T. (2014). Australia | Prospective cohort study of people aged ≥65 who had fallen and called for an ambulance (n=1610) | Collection of fall-related data allowing for a sequential description of fall-related cases from time of ambulance dispatch through to the end of the prehospital episode of care. | From 1,610 eligible cases and "long-lies" (>60 minutes on the ground) occurred in 13% of cases. Patients were predominantly female and community dwelling.<br><br>The most common patient-reported reasons for falling were:<br>Loss of balance<br>Simple trips<br>New injury and/or pain was documented for 1,172 of patients<br>656 presented with abnormal" physiology<br>Only 238 presented with no new injury/pain and normal physiology. | The paper concludes ambulance services appear to provide timely responses to older people who have fallen, and "long-lies" are relatively uncommon. |
| Scott, J (2020) | Retrospective cross-sectional cohort study in South Eastern division of the Northern Ireland Ambulance Service (NIAS). | Subsequent contacts with the ambulance service following referral. | There were 1079 incidents of falls in service users aged ≥ 65 years with a referral rate of 7% (n = 75) was reported. Re-contact rates were 37.3% (n = 28) within 1 month and 70.7% (n = 53) within 6 months.<br><br>Women and those experiencing a 'long lie' were more likely to re- | Interventions targeted at reducing long lies and supporting patients with cognitive impairments should be explored. Clinical documentation needs to be improved. |

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| | | | contact. Those with cognitive impairment were vulnerable to falls and repeat falls. Documentation for recording patient-related information was poor. | |

*Table 2: Table of included studies*

# Statements

## Contributorship

JB was the primary author. All authors made substantial contributions to the design, analysis and interpretation of this review. All authors were involved in drafting of the work and the development and review and revision of the final manuscript.

## Registration and protocol.

This review was not registered and a protocol was not prepared.


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## Competing interests

There are no competing interests.

## Data Sharing/Data availability

This manuscript does not use any patient identifiable information.

## Ethics Approval

Ethics Approval - Not Applicable

## PPI

No patients were involved